Sarvesh Sanjay Varadkar

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Education

D.Y. Patil College of Engineering, Akurdi, Pune, Maharashtra *June 2022 – Jan 2025*
Bachelor of Engineering — CGPA: 9.30

Technologies

Languages: Java, C++, Python.
Technologies and Frameworks: HTML , Flask , MongoDB. Gemini, Hugging Face, MySQL , Docker, Tensorflow ,Python , SpringBoot ,mlflow .
Deployment and Version Control: Git , GitHub, Vercel , DagsHub , DVC , Mlflow .
Competitive Programming: Code360, LeetCode 150+

Work Experience

Data Science Intern *[Feb to March]* **Nullclass Edtech Private Limited**

- Executed 3 real-time computer vision projects using **CNNs**, **YOLOv8**, and image processing for age, animal, and vehicle color detection. Built interactive GUIs with **PyQt5/Tkinter** for live video analysis, object classification, and event notifications. Applied transfer learning and custom object detection on datasets like **UTKFace** and **VCOR** to solve real-world challenges.

AI-ML Intern *[Jan to March]* **Eduskills Virtual Internship**

- Completed hands-on labs and earned 8+ Google Cloud badges, including: **TensorFlow**, **Object Detection**, **Image Classification**, **Product Image Search**
- Gained practical experience in building and deploying machine learning models using Google Cloud AI/ML tools.

Projects

Short Term Load Forecasting
Tech Stack: Tensorflow, Python, Flask , FastAPI , Pytorch , BeautifulSoup , Python , Flask , REST API, Pandas

- Designed and implemented an **LSTM neural network** model using **TensorFlow** and Keras to forecast hourly **electricity load** patterns in **time series data** with high accuracy.
- Applied **sequence reshaping, sliding window techniques, and model tuning** (epochs, batch size, neurons) to optimize prediction performance.

JournalApp
Tech Stack: SpringBoot, MongoDB, REST APIs, *Firebase*, Role Based Authentication

- Developed a secure journaling web application that allows users to create, update, and delete personal journal entries. Implemented **role-based access control** using **Spring Security** so that users can only access their own data.
- Built **RESTful APIs** with robust **error handling** and integrated **MongoDB** for persistent storage with user-specific document retrieval.

RAGnition: Multimodal Retrieval-Augmented Generation(RAG) System
Tech Stack: LangChain ,Groq (LLM) ,ChromaDB ,Streamlit ,LangSmith ,Python ,Huggingface Embeddings

- Built RAGnition, a multimodal RAG system using LangChain, Groq, and ChromaDB to enable AI chat over PDFs, images, URLs, and youtube videos.
- Engineered optimized pipelines for chunking, embedding, and retrieval, ensuring high-accuracy contextual responses and scalable vector store integration.

Positions of Responsibility and Achievements

- **AI-ML Lead at GDGC DYPCOE** - Guided students in their projects and conducted multiple workshops.
- **ACES Tech Lead** - Leading the technical team in organizing events and workshops.
- **Smart India Hackathon** 2024 Finalist (Runner Up + Top 5 India)
- Winner of the **Avishkar Competition** SPPU (Consecutively 3 Years)
- Completed course on Udemy on Data Science Bootcamp.